



CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 24: Measures of Central Tendency (Mean, Median, Quartiles and Mode)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Mean of raw and tabulated data

1. Assertion (A): The mean of the first 9 natural numbers is 4.5. Reason (R): Mean = $\frac{\text{sum of all the observations}}{\text{total number of observations}}$. (a) A true, R false (b) A false, R true (c) both true, and R is the correct reason for A (d) both true, but R is not the correct reason for A **ICSE 2025, 1 mark**

SOLUTION

$1 + 2 + \dots + 9 = 45$, so the mean is $\frac{45}{9} = 5$, not 4.5. A is false.

R is the correct formula for the mean.

Answer: (b)

2. For the data below, find (a) the mean, to the nearest whole number (b) the median.

ICSE 2025, 3 marks

x	10	11	12	13	14	15	16
f	3	2	2	6	3	5	3

SOLUTION

(a) $\sum f = 24$ and $\sum fx = 30 + 22 + 24 + 78 + 42 + 75 + 48 = 319$. Mean = $\frac{319}{24} \approx 13.29$, which is 13 to the nearest whole number.

(b) Cumulative frequencies: 3, 5, 7, 13, 16, 21, 24. With $n = 24$ (even), the median is the mean of the 12th and 13th values.

Values 8 to 13 are all 13, so the 12th and 13th values are both 13.

Answer: (a) mean = 13 (b) median = 13

3. The mean of the data below is 16. Find f . **ICSE 2020, 3 marks**

Marks	5	10	15	20	25
Number of students	3	7	f	9	6

SOLUTION

$$\sum f = 25 + f \text{ and } \sum fx = 15 + 70 + 15f + 180 + 150 = 415 + 15f$$

$$\frac{415 + 15f}{25 + f} = 16, \text{ so } 415 + 15f = 400 + 16f$$

Answer: $f = 15$

2 Mean of grouped data

4. The table gives the running times of some films. Find the mean running time by the step-deviation method. **ICSE 2024, 3 marks**

Duration (minutes)	100–110	110–120	120–130	130–140	140–150	150–160
Number of films	5	10	17	8	6	4

SOLUTION

Class marks: 105, 115, 125, 135, 145, 155. Take $A = 125$ and $h = 10$, so $u = \frac{x - 125}{10}$ is

$-2, -1, 0, 1, 2, 3$.

$$\sum f = 50 \text{ and } \sum fu = -10 - 10 + 0 + 8 + 12 + 12 = 12$$

$$\text{Mean} = A + \frac{\sum fu}{\sum f} \times h = 125 + \frac{12}{50} \times 10$$

Answer: 127.4 minutes

5. Find the mean of the marks below by the short-cut method. **ICSE 2023, 3 marks**

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	3	8	14	9	4	2

SOLUTION

Class marks: 5, 15, 25, 35, 45, 55. Take the assumed mean $A = 25$, so $d = x - 25$ is $-20, -10, 0, 10, 20, 30$.

$$\sum f = 40 \text{ and } \sum fd = -60 - 80 + 0 + 90 + 80 + 60 = 90$$

$$\text{Mean} = A + \frac{\sum fd}{\sum f} = 25 + \frac{90}{40}$$

Answer: 27.25 marks



3 Median of raw data

6. The numbers 27, 31, 46, 52, x , $x + 4$, 71, 79, 85, 90 are in ascending order and their median is 64. Find x . (a) 60 (b) 61 (c) 62 (d) 66 **ICSE 2024, 1 mark**

SOLUTION

There are 10 numbers, so the median is the mean of the 5th and 6th: $\frac{x + (x + 4)}{2} = x + 2$.

$x + 2 = 64$, so $x = 62$.

Answer: (c)

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